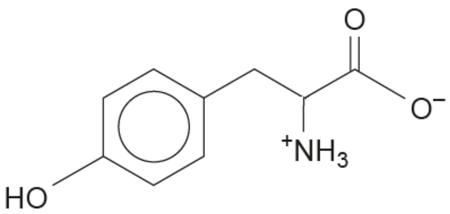
Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)	(i)		chromatogram drawn correctly with spot at 6 cm mark		1		1		1
		(ii)			1			1		
		(iii)		the polar structure / OH group is a small part of the overall molecule so hydrogen bonding is at a 'minimum'			1	1		
		(iv)	I	24500 cm^3 of nitrogen from 181 g of tyrosine (1) 1 cm^3 of nitrogen from $\frac{181}{24500}$ g of tyrosine 147 cm^3 of nitrogen from 147 × $\frac{181}{24500}$ = 1.09 g(1) accept alternative methods e.g. n = $\frac{pV}{RT}$ = 0.006 (1) mass = 0.006 × 181 = 1.09g (1) e.g. 1 mol tyrosine gives 1 mol N_2 n(N_2) = $\frac{1.09}{181}$ = 6 × 10^{-3} (1) V = $\frac{nRT}{p}$ = 147cm^3 (1)		2		2	1	